

ALL ABOUT  
WATER  
TESTING  
PARAMETERS  
AND METHODS  
IN TMC VARTAK  
NAGAR LAB

- I. **Dissolved Oxygen (DO):** The amount of oxygen available for aquatic life. DO as a limit of 4.0
- II. **Chemical Oxygen demand (COD):** Oxygen consumed by chemical to break down both biodegradable and non-biodegradable organic matter. To find the chemical quantity by finding the oxygen demand. COD as a limit of 50 mg per liter
- III. **Biological Oxygen demand (BOD):** Oxygen consumed by microbes to break down biodegradable organic matter. To find the microbes quantity by finding the oxygen demand. BOD as a limit of 10 mg per liter
- IV. **Turbidity:** Measures the cloudiness or haziness of water caused by suspended particles. Limit is not consider.
- V. **Suspended Solids (SS):** Measures the amount of solid particles in a water sample that are not dissolved and are trapped by a filter. SS as a limit of 100 mg per liter
- VI. **NH<sub>4</sub>:** It is a Test of ammonia present in water. NH<sub>4</sub> as a limit of 1.2.
- VII. **PH:** To water is acidic or base or neutral water. PH as a limit of 6.5 – 8.5
- VIII. **Total nitrogen :** To seen Total nitrogen (N) like nitrate (N<sub>2</sub>) and nitride (N<sub>3</sub>) and ammonia (NH<sub>4</sub>) In water. 1.2 mg per liter is a limit
- IX. **Hardness:** The concentration of dissolved minerals, primarily calcium and magnesium. Limit is not consider
- X. **Total Dissolved solids (TDS):** Measures the total amount of dissolved substances in water. Limit is not consider.
- XI. **hazards heavy matel in water are also check in water testing parameter**

| LAKE      | SWIMMING<br>POOL | CONSTRUCTION            | SEWAGE<br>TREATMENT<br>PLANT<br>( untreated) | SEWAGE<br>TREATMENT<br>PLANT<br>(treated) | Amrut lake        |
|-----------|------------------|-------------------------|----------------------------------------------|-------------------------------------------|-------------------|
| PH        | PH               | PH                      | PH                                           | PH                                        | PH                |
| DO        | DO               | DO                      | COD                                          | COD                                       | DO                |
| TDS       | TDS              | TDS                     | BOD                                          | BOD                                       | TDS               |
| Ammonia   | SS               | Acidity                 | SS                                           | SS                                        | Ammonia           |
| Turbidity | BOD              | Basicity or<br>Alkaline | DO                                           | DO                                        | Total<br>nitrogen |
|           |                  | COD                     | TDS                                          | TDS                                       | Turbidity         |
|           |                  | BOD                     | Ammonia                                      |                                           | COD               |
|           |                  | Sulphate                | Total<br>nitrogen                            |                                           | BOD               |
|           |                  | Chloride                |                                              |                                           | MPN               |
|           |                  | Iron                    |                                              |                                           |                   |

Note:

Amrut Yojana is a central funded scheme to rejuvenation bodies like lakes in Maharashtra. 15 lakes are currently under Amrut yojana lakes. Kharegaon, Hariyali, Shivajinagar-Kalwa, Kausa, Kolshet , Dativali, Desai, Brahma (also spelled Bramhala), Ambe-Ghosale, Kacharli (also spelled Kachrali), Kamal, Khidkali (also spelled Kidkali), Jogila, Railiadevi, tursepada.

Distilled water :



Distilled water is a highly purified form of water created by boiling water into vapor and condensing it back into a liquid, leaving behind contaminants, minerals, and impurities. To Production it take boiled Water, and the resulting steam is condensed in a separate container, separating it from contaminants, bacteria, and minerals. It is nearly 100% pure water ( $H_2O$ ). It is safe to drink but has a flat or bland taste due to the removal of minerals like calcium, sodium, and magnesium. Distilled water is theoretically neutral with a pH of 7.0, but in practice, it is usually slightly acidic, typically ranging between 5.5 and 6.9 because of absorb Carbon Dioxide near it.

Globally Harmonized System:

GHS Hazard pictograms:

| pictograms                         | Symbol                                                                                                                                                                                                                             | Meaning                                                                                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute toxicity<br>(GHS06)          |  A black skull and crossbones symbol inside a red diamond border.                                                                                 | Indicates that the products is potentially fatal, toxic, or harmful if swallowed, inhaled, or comes into contact with skin. It typically warns of severe, immediate health effect even at low doses. |
| Health Hazards<br>(GHS07)          |  A black silhouette of a person with a starburst symbol on the chest, indicating a health hazard, inside a red diamond border.                   | Indicates hazards such as skin and eye irritation, skin sensitize, and sometimes acute toxicity (harmful)                                                                                            |
| Corrosion<br>(GHS05)               |  A black symbol showing two test tubes pouring liquid onto a surface and a hand, with flames indicating corrosion, inside a red diamond border. | Indicates the product can cause severe skin burns and eye damage upon contact. It also signifies that the substances is corrosive to metals.                                                         |
| Environmental<br>Hazard<br>(GHS09) |  A black symbol showing a dead tree and a dead fish, indicating environmental harm, inside a red diamond border.                                | Indicates that the product is an environment hazard and is toxic to aquatic life (fish, organisms, etc.) or causes long-term harmful effects in the environment.                                     |

## Spectro Photo meter:



In the given above images is a spectro photo meter, It measures a range of parameters are Suspended Solids (SS) and Turbidity parameters. it is a visible-range spectrophotometer commonly used in water and wastewater analysis. Light passes through the water sample. The machine checks how much light is absorbed by the sample. Different substances absorb light differently, so the machine calculates the concentration of that substance. It then shows the result on the screen (like mg/L). It use cuvette (you can see a cuvette placed on the instrument in the image) Fill water sample in it to test.

## Spectro Photo meter:



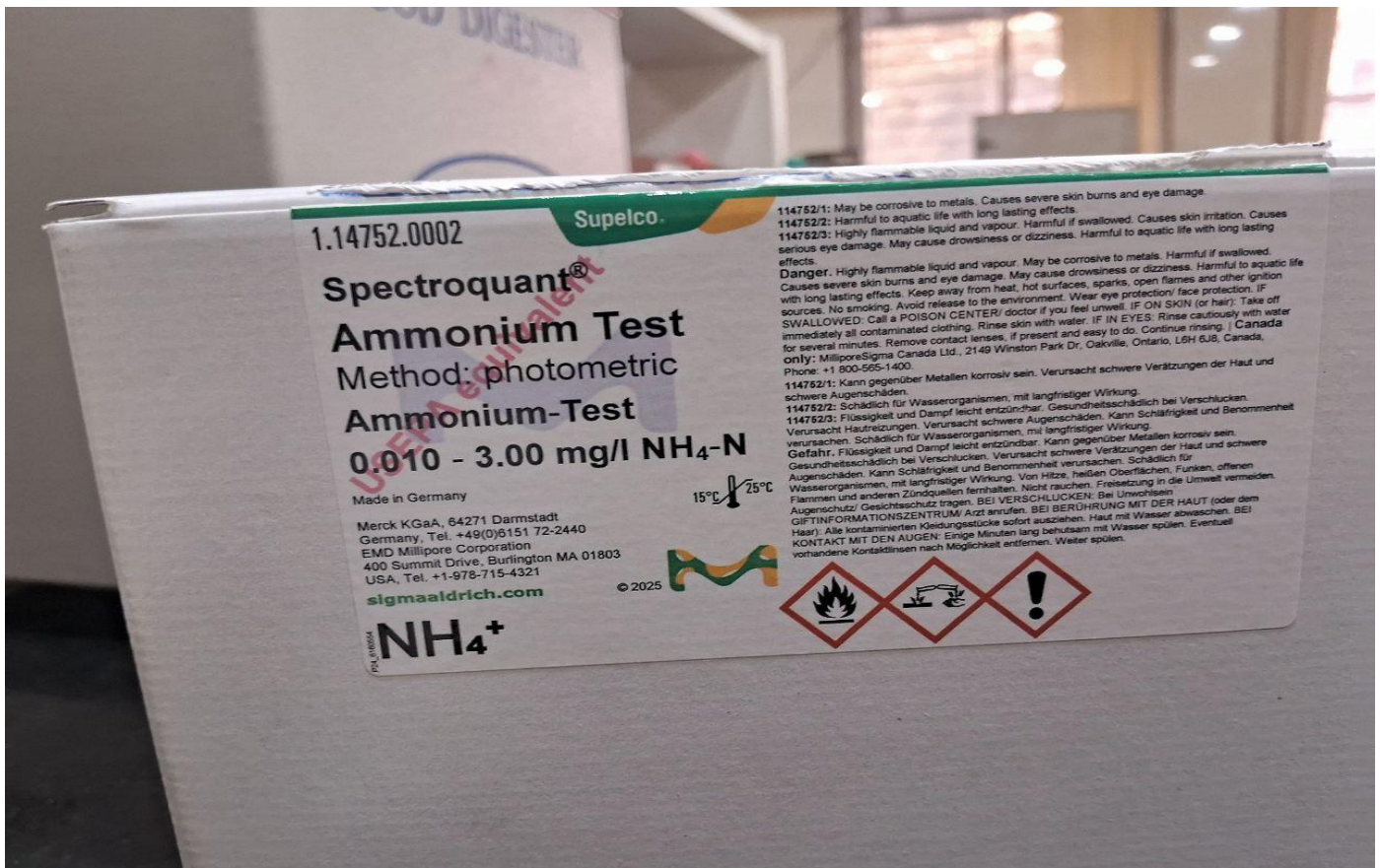
In the given above images is also a spectro photo meter, we measures a range of parameters are Chemical Oxygen demand (COD), total nitrogen and amonin( $\text{NH}_4$ ) parameters. is a UV–Visible (UV-Vis) spectrophotometer, a laboratory instrument used to measure how much light a chemical sample absorbs at different wavelengths. It is advance then the above spectro photo meter given. A light source emits UV and visible light. The light passes through a sample in a cuvette (you can see a cuvette placed on the instrument in the image). Depending on what chemicals are in the sample, certain wavelengths get absorbed.

## COD digester:



In the given image is a COD digester, A COD Digester is a heating block used for digesting water samples during COD analysis. COD (Chemical Oxygen Demand) measures how much organic matter is present in water. To determine this, the sample must be heated with strong chemicals (potassium dichromate + sulfuric acid) at  $150^{\circ}\text{C}$  for 2 hours, Ensures complete oxidation of organic matter in the sample. This two digester are also help in at digestion of total nitrogen method. The left one is for big quantity and right one is for small quantity of testing parameters.

## Ammonium Test kit :



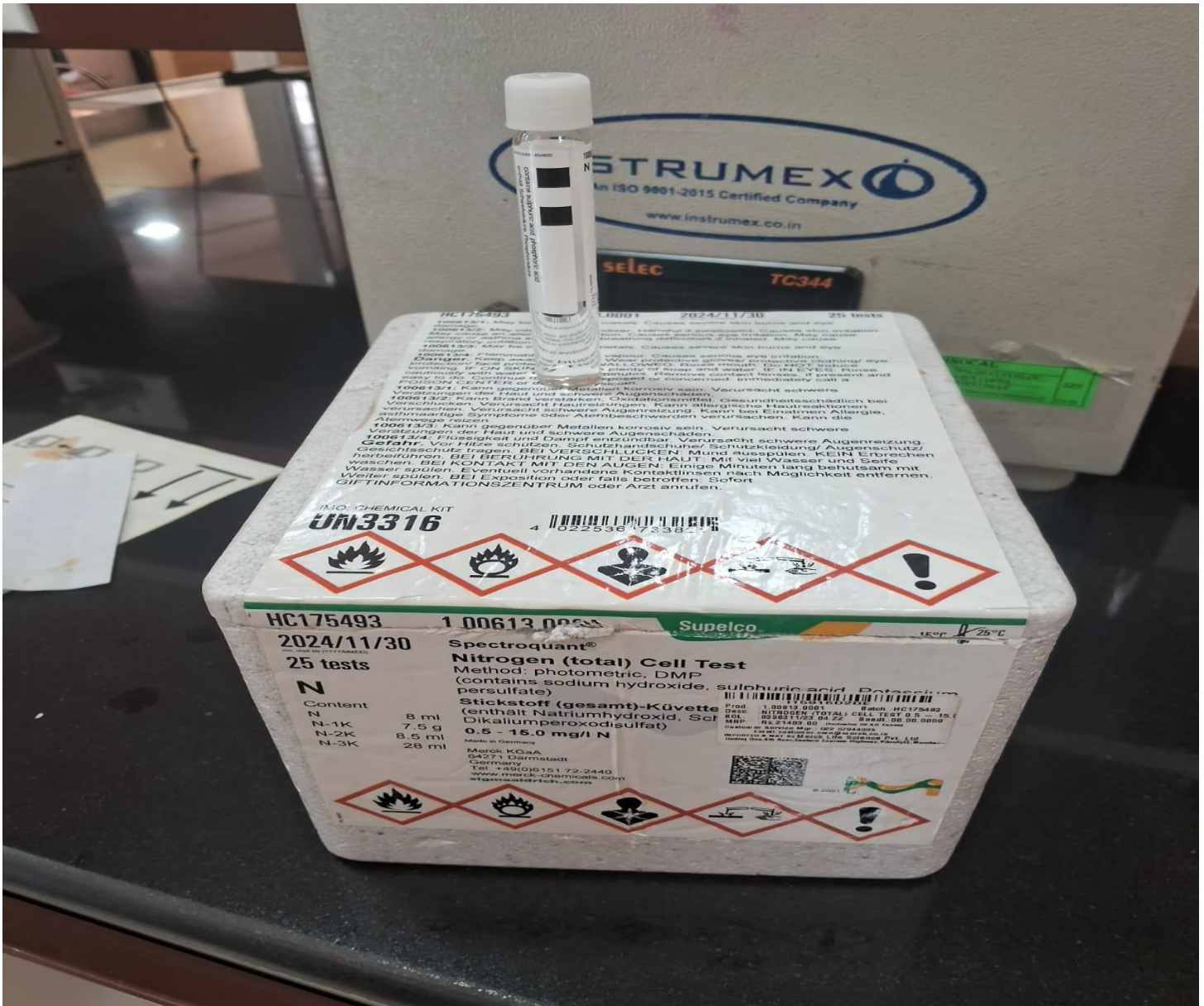
It is an ammonium testing kit, specifically for the lakes. Indicator of fresh pollution. Toxic to aquatic life at higher concentrations. Affects drinking water taste and safety. Important parameter in water and wastewater analysis. Ammonium ions react with reagents to form a colored complex. The intensity of the color formed is proportional to the ammonium concentration. The absorbance is measured using a spectrophotometer. The instrument converts absorbance into NH<sub>4</sub>-N concentration (mg/L). The Spectroquant Ammonium Test is a precise and reliable photometric method for determining ammonium levels in water. It is widely used for pollution monitoring, water quality assessment, and environmental studies, especially for detecting fresh nitrogen contamination.

COD kit :



In this given images is Chemical Oxygen Demand (COD) kit, Widely used in environmental monitoring, wastewater treatment, and industrial effluent analysis. this test is done for measures the amount of oxygen required to chemically oxidize organic and inorganic matter in water serving as a key indicator of pollution levels. It is primarily used to assess wastewater quality, monitor treatment plant efficiency, and determine pollution loads in industrial and municipal effluents. It is done with a already ready made sample like shown in the image of on the box it is  $\text{K}_2\text{Cr}_2\text{O}_7$  that mix with measured sample of lake about 2ml and then put it in digester for 2 hours in 120 to 150°C for a complete chemical reaction of water sample and the oxidizing reagent, The resulting color change is measured using a spectrophotometer. Absorbance is directly proportional to COD concentration. Also Chemical Oxygen demand (COD) should not be more than Biological Oxygen demand (BOD).

Total nitrogen :



In the given image is a total nitrogen kit, Total Nitrogen (TN) represents the sum of all nitrogen compounds present in a water sample, including: Ammonium ( $\text{NH}_4^+-\text{N}$ ), Nitrate ( $\text{NO}_3^--\text{N}$ ), Nitrite ( $\text{NO}_2^--\text{N}$ ).key measure of water pollution and a primary indicator , where excessive nutrients trigger harmful algal blooms that deplete oxygen and threaten aquatic life. It is a critical analytical parameter used in environmental monitoring, wastewater management, agriculture, and the food industry to assess nutrient levels, pollution, and product quality. It is done with a already ready made sample like shown in the image of on the box it is Potassium peroxodisulfate that mix with measured sample of lake about 2ml and then put it in digester for 2 hours in 120 to 150°C for ensure all nitrogen forms are converted into all three total nitrogen. After digestion, nitrate reacts with specific reagents to form a colored compound. The intensity of the color is measured using a spectrophotometer. The absorbance is directly proportional to the total nitrogen concentration. Importance of test is to Indicates pollution,Excess nitrogen causes eutrophication.

## PH Meter:



In this image It is a Digital pH Meter instead of chemical we use the Electrometric. This setup is used to measure the pH of water or liquid samples accurately in laboratory conditions. The glass electrode develops an electric potential when immersed in solution. This potential depends on hydrogen ion concentration ( $H^+$ ). A reference electrode provides a stable comparison voltage. The meter measures the potential difference. This voltage is converted into a pH value and displayed digitally. pH is a measure of the acidity or alkalinity of a solution. pH scale ranges from 0 to 14 :

$pH < 7 \rightarrow$  Acidic

$pH = 7 \rightarrow$  Neutral

$pH > 7 \rightarrow$  Alkaline (Basic)

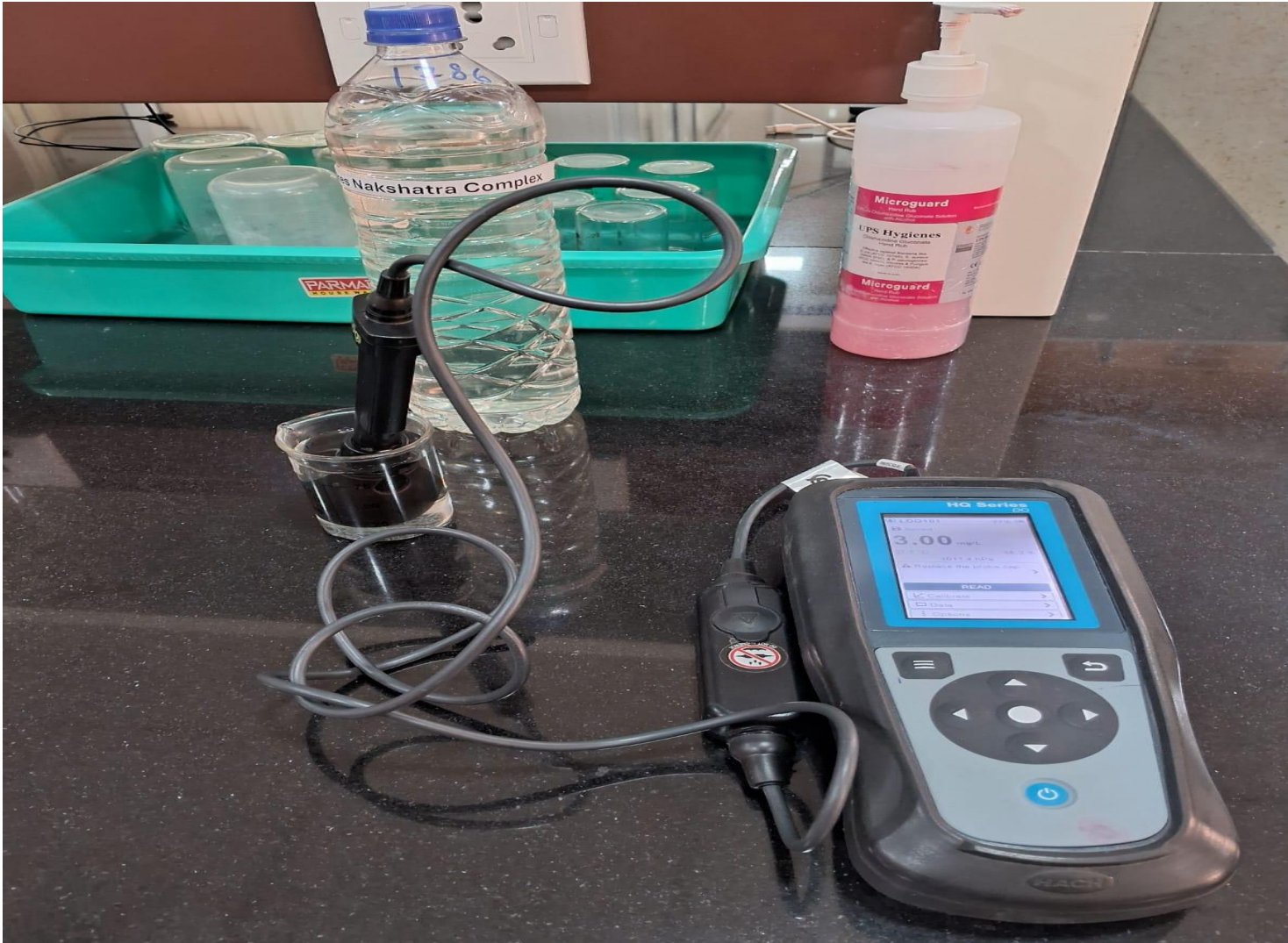
The observed pH value of 8.05 indicates a mildly alkaline sample, suitable for environmental and water quality studies.

## TDS Meter:



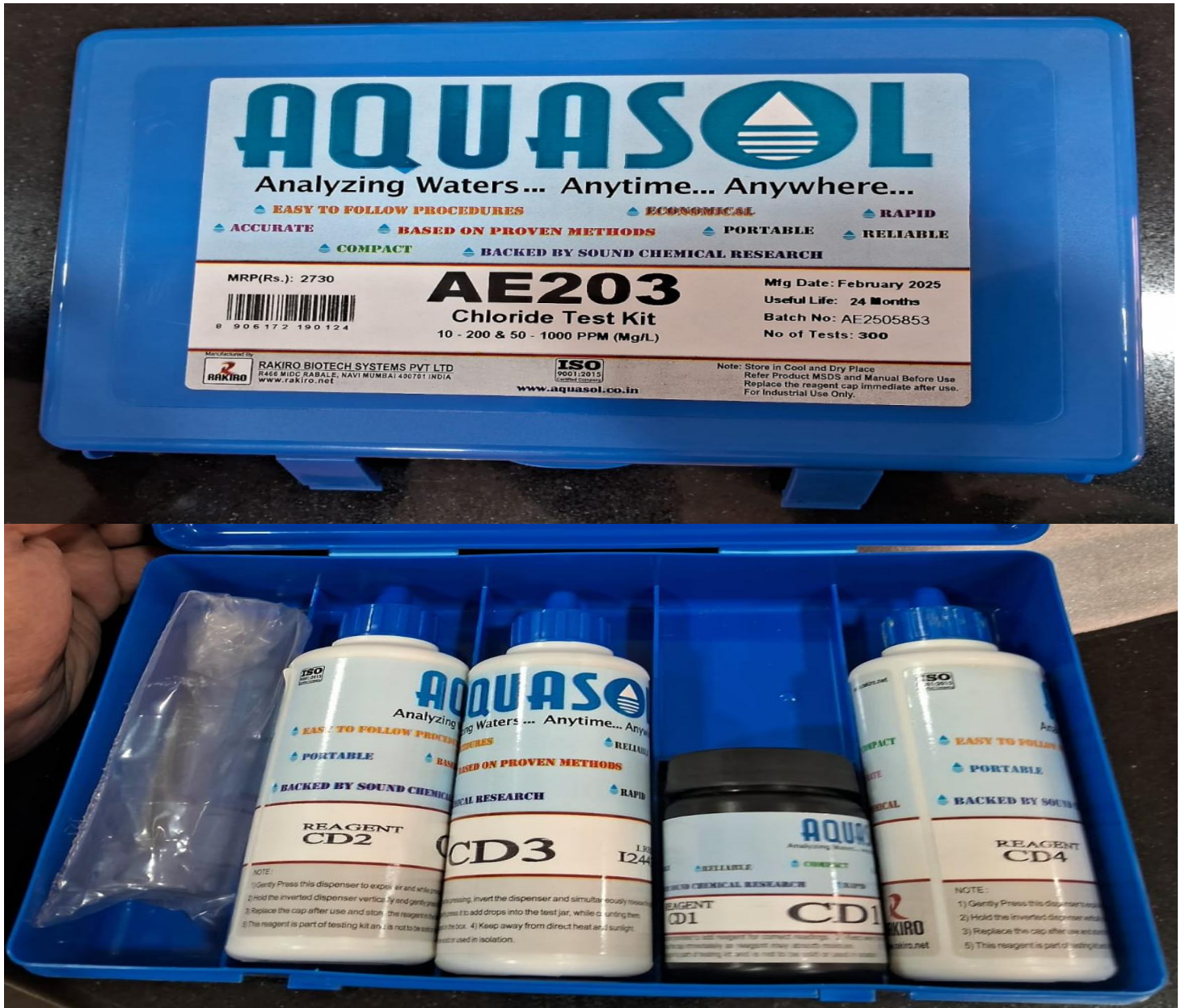
In a above image it is a Instrument called TDS meter, it is a Portable electronic instrument. To measure the Total dissolve solid simply dip the TDS meter in water in such a such a way that the two metal tips are completely submerged in the water as shown in above image. If the sample TDS run out of limit (above 1500) and show nothing then sample will dilute in 10 ml of Distilled water and again measure in TDS meter then the result should be multiplied by 10. The digital TDS meter is a simple and effective instrument for determining total dissolved solids in water.

## DO Meter:



In this given image is a DISSOLVED OXYGEN (DO) METER, It is also a portable but heavier than TDS meter. Built into the probe. Provides automatic temperature compensation. Important because DO solubility changes with temperature. Probe is immersed in the water sample Oxygen diffuses through the probe membrane Chemical reaction generates electrical current is converted into DO concentration Result is displayed digitally on the screen. It is Fast and accurate results, Portable and easy to operate, Suitable for field and lab use, Digital display reduces human error. A Dissolved Oxygen meter is an essential instrument in environmental science used to assess water quality and aquatic health. It provides rapid, accurate measurement of oxygen levels and helps in detecting pollution and ecosystem stress.

## Chloride Test kit:



A chloride test kit is used to measure the concentration of chloride ions in various samples, including water (drinking, industrial, wastewater). High chloride content in industrial water, such as in boilers and cooling systems, can lead to the corrosion and damage of metal pipes and equipment. Easy to use, Portable and compact, Rapid results, Field-friendly No electricity required, Based on standard chemical method. The AQUASOL Chloride Test Kit (AE203) is a reliable and portable analytical tool for estimating chloride content in water using the argentometric titration method. It is widely used for environmental monitoring, water quality assessment, and educational laboratory experiment.

## Total Hardness kit:



Total hardness of water is a measure of the concentration of calcium ( $\text{Ca}^{2+}$ ) and magnesium ( $\text{Mg}^{2+}$ ) ions present in water. Determines suitability for drinking, Prevents scaling in pipelines, Protects boilers and heat exchange and Important for water treatment processes. A total hardness test kit is used to measure the concentration of dissolved minerals in water, primarily calcium and magnesium ions, which are the main contributors to water hardness. The primary use for these kits is to monitor water quality in industrial systems like boilers, cooling towers, and plumbing. Hard water leads to scale deposits (limescale) which reduce efficiency, increase energy consumption, and damage equipment over time. The AQUASOL Total Hardness Test Kit (AE201) is a reliable and portable analytical tool used to determine the total hardness of water based on the EDTA titration method. It is widely used in water quality assessment, environmental monitoring, and educational laboratories.